

Computer Science & IT

Discrete Mathematics

Mathematical Logic

Lecture No. 01

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Recap of Previous Lecture



Topic

Clique

Topic

Practice questions



Topics to be Covered



✓ **Topic**

Propositions and their types

✓ **Topic**

Connectives

✓ **Topic**

Tautology, Contradiction, Contingency and Satisfiable propositional functions

Mathematical Logic

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graph TD; A[Mathematical Logic] --> B[① Propositional Logic]; A --> C[② Predicate Logic];
```

✓
① Propositional Logic

② Predicate Logic
X



Topic : Propositional Logic

* Proposition : —

A declarative sentence to which we can assign only one of the truth values (i.e. TRUE/FALSE) is called a propositional statement or proposition.

- eg:
- It rains today
 - Cricket match is played.



Topic : Assumptions of proposition



Law of excluded middle : If a proposition is not true then it will be false, and similarly if a proposition is not false then it will be true.

Law of contradiction : A proposition can not be true as well as false simultaneously.



Topic : Atomic propositions



Atomic propositions are simple propositions which can not be divided further.

eg: ① It rains today

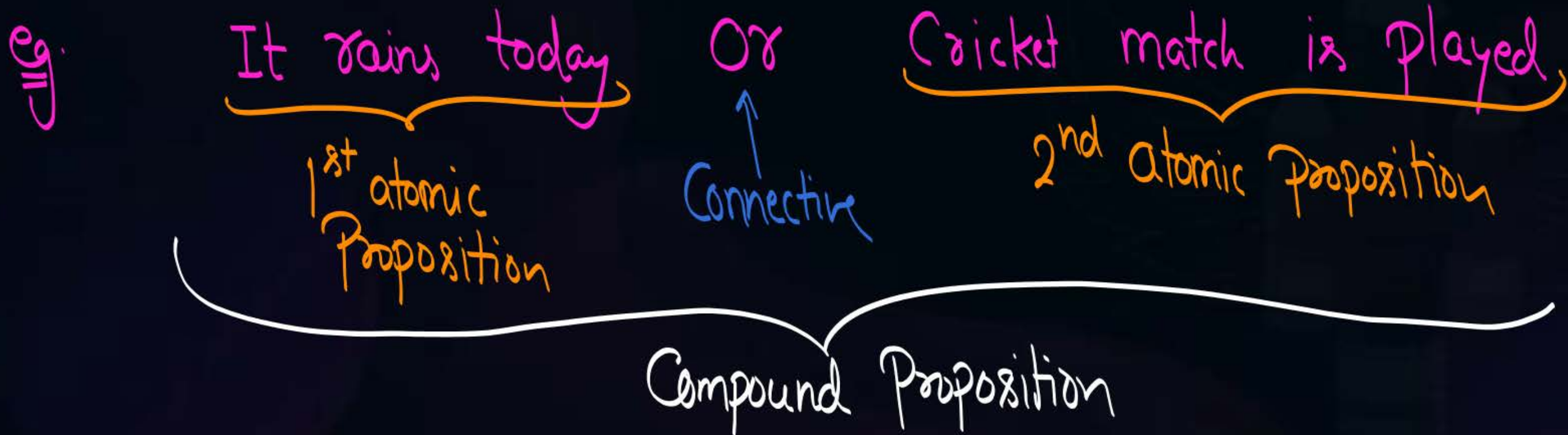
② Cricket match is played.



Topic : Compound Propositions

- Compound propositions are constructed by combining atomic propositions with the help of connectives.

Compound propositions are also known as propositional formula, or propositional function.





Topic : Connectives

- ✓ 1. Negation ($\neg/\sim$) / Not
- ✓ 2. AND ($\wedge$) / Conjunction
- ✓ 3. OR ($\vee$) / Disjunction
- ✓ 4. Implication ($\rightarrow$)
- ✓ 5. Biconditional ($\leftrightarrow$)

Imp.



Topic : Negation ($\neg/\sim$)

- If p is any proposition, then negation of p / not(p) / $\neg p$ is also a proposition whose truth value is true only when p is false and it is false only when p is true.

p	$\sim p$
T	F
F	T



Topic : AND ($\wedge$) / Conjunction

If p and q are any two propositions, then “ p AND q ” / $p \wedge q$ is also a proposition whose truth value is true only when both p as well as q are true.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F



Topic : OR ($\vee$) / Disjunction

If p and q are any two propositions, then " p OR q " / $p \vee q$ is also a proposition whose truth value is false only when both p as well as q are false.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

If it rains today then Cricket match will not be played

p q

if p then q

p	q	$p \rightarrow q$
1	1	1
1	0	0
0	1	1
0	0	1



Topic : Implication ($\rightarrow$)

aka (if-then)

If p and q are any two propositions, then p implies q (i.e. $p \rightarrow q$ / if p then q) is also a proposition whose truth value is false only when p is true, and q is false.

$p \rightarrow q$
Antecedent Consequent

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

" $P \rightarrow Q$ " & " $\sim P \vee Q$,"
have same truth table
∴ $P \rightarrow Q \equiv \sim P \vee Q$

We can observe
 $Q \rightarrow P \equiv \sim P \rightarrow \sim Q$

P	Q	$\sim P$	$\sim Q$	$P \rightarrow Q$	$\sim P \vee Q$	$Q \rightarrow P$	$\sim P \rightarrow \sim Q$	$\sim Q \rightarrow \sim P$
T	T	F	F	T	T	T	T	T
T	F	F	T	F	F	F	F	F
F	T	T	F	T	T	T	T	F
F	F	T	T	T	T	T	T	T

$P \rightarrow Q \equiv \sim Q \rightarrow \sim P$



Topic : Implication ($\rightarrow$)

- ❑ If p is true, then q must be true for $p \rightarrow q$ to be true.
- ❑ If p is false, then $p \rightarrow q$ is *always* true whatever is the truth value of q .
- ❑ If q is true, the $p \rightarrow q$ is *always* true whatever is the truth value of p .
- ❑ Truth table of $p \rightarrow q$ and truth table of $\neg p \vee q$ are exactly same, therefore both are equivalent.
- ❑ Converse of $p \rightarrow q$ is $q \rightarrow p$.
- ❑ Opposite or Inverse of $p \rightarrow q$ is " $\neg p \rightarrow \neg q$ "
- ❑ Contra positive for $p \rightarrow q$ is " $\neg q \rightarrow \neg p$ "

$$p \rightarrow q \equiv \sim p \vee q$$

$$p \rightarrow q \equiv \underbrace{\sim q \rightarrow \sim p}_{\text{Contra positive of } p \rightarrow q}$$

$$p \rightarrow q \not\equiv \sim p \rightarrow \sim q$$

(inverse of $p \rightarrow q$)

$$p \rightarrow q \not\equiv q \rightarrow p$$

(converse of $p \rightarrow q$)

$$q \rightarrow p \equiv \sim p \rightarrow \sim q$$

Converse of $p \rightarrow q$ is
equivalent to inverse of $p \rightarrow q$



Topic : Biconditional ($\leftrightarrow$)

(if and only if)

- If p and q are any two propositions, then " $p \leftrightarrow q$ " (i.e. p if and only if q) is also a proposition whose truth value is true only when both p and q have same truth value.

p	q	$p \leftrightarrow q$	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T	T
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

We can observe
 $(p \leftrightarrow q) \equiv (p \rightarrow q) \wedge (q \rightarrow p)$



Topic : Biconditional ($\leftrightarrow$)

We observed $(p \leftrightarrow q) \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

" $p \leftrightarrow q$ " is true if and only if
both " $p \rightarrow q$ " and " $q \rightarrow p$ " are true



Topic : Tautology

{ Tautology is term which is defined w.r.t. propositional logic }



➤ A propositional function which is always true is called tautology.

A propositional function which is always true is also called a valid propositional function.

W.r.t. propositional logic,

Tautologies & Valid propositional function
are same things

{ But it will not be the case w.r.t. predicate logic }

eg.

$$p \vee \sim p \quad f(p)$$

p	$\sim p$	$p \vee \sim p$
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T

F

T

F

T

T

} Always true
∴ Tautology

eg.

Check if $p \rightarrow (q \rightarrow p)$ is a tautology or not

p	q	$(q \rightarrow p)$	$p \rightarrow (q \rightarrow p)$
T	T	T	T
T	F	T	T
F	T	F	T
F	F	T	T

} Always True

∴ $p \rightarrow (q \rightarrow p)$ is a tautology



Topic : Contradiction



- A propositional function which is always false is called Contradiction.

$f(p)$

p	$\sim p$	$p \wedge \sim p$
T	F	F
F	T	F

Always false
∴ Contradiction

Note:-

① If a propositional function is always true then it is called Valid propositional function

② If propositional function is false for at least one case, then it is an invalid Propositional function

✶ Each Contradiction is invalid propositional function, but Each invalid propositional function need not be Contradiction



Topic : Contingency



A propositional function which is neither a tautology nor a contradiction is called a contingency.

i.e., A contingency is true for at least one case as well as false for at least one case.

eg:-

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

$f(p, q)$

True for at least one Combination
as well as

false for at least one Combination

so it is a Contingency.



Topic : Satisfiable



A propositional function which is true for at least one case is called a satisfiable propositional function.

- i.e., if a propositional function is not satisfiable, then it is a Contradiction
- if propositional function is not a Contradiction, then it is satisfiable



Topic : NOTE



- Every propositional function which is not a contradiction is satisfiable.
- Every tautology is satisfiable but not vice-versa.
i.e. Every Satisfiable propositional function, need not be a tautology
- Every contingency is satisfiable, but every satisfiable function need not be contingency.
→ eg: Tautology — it is Satisfiable, but not a Contingency.



Topic : NOTE



Tautology	Contradiction	Contingency
✓ Always True	✓ Always False	✓ Neither always true nor always false
✓ Valid	✓ Invalid	✓ Invalid
✓ Satisfiable	✓ Not satisfiable	✓ Satisfiable

H.W. Q. Consider the following propositional function
 $F: \{p \rightarrow (q \vee r)\} \rightarrow \{(p \wedge q) \rightarrow r\}$,
then propositional function F is

- (a) Contingency
- (b) Valid
- (c) Contradiction
- (d) Can not be defined



2 mins Summary



✓
Topic

Propositions and their types

✓
Topic

Connectives

✓
Topic

Tautology, Contradiction, Contingency and Satisfiable propositional functions

THANK - YOU